ICT302 IT Professional Practice Project

Project Name:

Secure QR Code Development

Client Organization:

Secure QR Code Project Client

Group Name:

VAFPQR

PROJECT MANAGEMENT PLAN

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1. Executive Summary （Patricia)

This document serves as the Project Management Plan (PMP) for the VAFPQR Project Team. This project delivers a secure QR code authentication platform designed to protect businesses and clients from the increasing threat of physical QR code tampering and fraud. The system combines cryptographic security, real-time monitoring, and automated tamper detection into a production-ready solution built for scalability and cross-industry adoption. The system integrates automatic tamper detection, real-time monitoring, and cryptographic security into a production ready solution designed for scalability and industry standard acceptance. Clients can generate and manage cryptographically secure QR codes, identify and report physical tampering attempts, and view scan analytics and tamper alarms via a centralized administration dashboard, among other end-to-end capabilities. When combined, these characteristics dramatically lower the risk of fraud while providing companies with real-time, transparent information about how their QR codes are being used in the field. The primary goal of the system is to address a real security gap that offers significant value to business and clients by generating secure QR codes.

2. Introduction (Patricia)

QR codes are now widely used in public signage, restaurant menus, payment terminals, and product packaging. However, its ease of use has created a serious, often overlooked security flaw. Fraudulent QR codes can be easily placed over or if not generated securely, can lead users to phishing pages, phony payment portals or harmful websites and these often come with no obvious warning that something is wrong. As QR code usage continues to grow across industries, so does the potential for exploitation.

This project directly addresses that gap. Businesses may create cryptographically secured QR codes, track their usage in real time, and receive automated notifications when tampering is discovered using the Secure QR Code Authentication System, a multi-layered platform. The system consists of three integrated parts, a web-based management dashboard that provides businesses with complete visibility into scan activity and security events, a mobile application that allows end users to scan and authenticate QR codes, and a backend API that handles code generation, validation, and data management.

With an emphasis on providing a production-ready system that is safe, scalable, and flexible across a variety of industries and use cases, the platform was created in close collaboration with clients Peter Cole and Mike Groeneweg. The design, architecture, development process, and results of the system are described in the document that follows.

3. Project Integration Management (Felixia)

Project Management Approach

The meeting for the project initiation was conducted on 20 May 2026. The team chat on WhatsApp was created on that same day. Submission of the Team Charter was made on 23 May 2026. In the first meeting, responsibility among team members was allocated, common OneDrive account was created, and guidelines on communication was also defined.

Team conducts online meeting once every week on either Thursday or Friday night. All team meetings take up a period of about two to three hours. In every meeting, each team member gives three updates which include what they have accomplished since the previous meeting, what plans they have for the next week, and what problems or challenges they are facing. Meeting minutes are also documented in every meeting and stored in the common OneDrive account.

The Task Board is used to record all the tasks done in the project. It comprises of four different sections namely To Do, In Progress, Completed, and Blocked. All the tasks listed in the work breakdown structure appear on the task board. The deadlines for each of the tasks along with the team member assigned are recorded with it. Project manager, Patricia evaluates the task board on two different occasions each week in order to see which team members are under-performing.

Both Peter and Mike get updates from our project team after each milestone is reached. The list of the main milestones includes: approval of requirements, approval of architecture, first demo, system ready for testing, and delivery of the product. Each of these milestones receives an email from the developers and a notification on the Microsoft Teams channel. The access to the development environment is given to the client so that they can view the system directly. A video demonstration may be made if necessary to show new system features.

At the end of the process, the customer will receive all code and documentation in PDF form as well as a one-hour training session delivered through Microsoft Teams. This training is focused on QR codes generation and system accounts management.

Change Management

Changes are expected in any project. A three-tiered change control process has been established.

Small changes having no effects on the projects can also be done after notifying the team and recording them in the change log. In such an instance, it will mean that the team member will have to notify other team members about the change using WhatsApp within 24 hours. Examples of changes that follow this process include fixing any errors in spelling and button color change.

Modifications requiring between two and eight hours of work will involve a voting process by the whole team in the weekly meeting. Here, any team member proposes the change, discusses its impact, and gets a unanimous vote. This will involve updating the task board and modifying the relevant due dates. Minor modifications will not require client approval unless it involves a change in system behavior or functionality.

For major modifications, involving more than eight hours of work or system functionality, the client's approval is required. This involves sending an email to Peter and Mike explaining why the client requested such a modification and the number of additional hours/days and dropping/postponing of any existing feature. The client will then give his approval via Microsoft Teams or email.

This process will help address Risk R01 (scope creep) and Risk R11 (schedule delay) according to the project risks register.

There are certain elements that never change no matter what without the approval from the client. The unchangeable elements include JWT signing procedure for the QR codes, blacklist for the compromised codes, tampering alert procedure, database design once coding is initiated, and the submission deadline date.

Project Management Methodology

In our case, the agile approach with the time period of two weeks is adopted. Such an approach can be highly efficient for the team of six. The total duration of the project from beginning till completion falls between 20th May 2026 and 1st August 2026.

Sprint	Weeks	Dates	Sprint Focus
Sprint 1	Week 1-2	20 May to 5 June	Project Management Plan, initial requirements gathering, team setup
Sprint 2	Week 3-4	6 June to 19 June	Requirements document finalisation, database design, backend setup
Sprint 3	Week 5-6	20 June to 3 July	Design Document completion, backend setup, QR generation endpoint, verification endpoint
Sprint 4	Week 7-8	4 July to 17 July	QR Code Landing Page, Admin dashboard pages, mobile scanner screen, tamper alert reporting
Sprint 5	Week 9-10	18 July to 31 July	Full system testing, bug resolution, documentation writing
Sprint 6	Week 11	1 August	Final submission of all deliverables, presentation, demo media, promo video

The following are two key activities that are conducted at the end of every sprint cycle. First, there is an exercise of providing feedback to the client via Microsoft Teams about what has been achieved through the sprint cycle and allowing the client to see the application that has been developed. At the end of each sprint, a retrospective exercise is carried out, where team members must respond to two major questions.

4. Project Scope (Roy Hojin)

4.1.1 Project Goals

Academic Goals

- Apply real-world software development practices in a team environment, including requirements gathering, system architecture design, iterative development, and testing.
- Demonstrate competency in full-stack development, mobile development, and secure system design.
- Produce professional-grade documentation and deliverables that reflect industry standards.

Business Goals

- Deliver a secure QR code authentication system that protects businesses and the public from physical QR code tampering and fraud.
- Provide clients Peter Cole and Mike Groeneweg with a production-ready platform that is scalable, maintainable, and adoptable across industries.

4.1.2 Project Benefits

Team Benefits

- Gain hands-on experience building a multi-layered system spanning backend API development, mobile application development, and web-based dashboard design.
- Develop professional collaboration skills including GitHub version control, task management, and client communication.
- Build a portfolio-ready project demonstrating real-world technical competency.

Client Benefits

- A complete platform for generating, managing, and authenticating QR codes with cryptographic security.
- Automated detection and reporting of physical QR code tampering, reducing fraud risk for businesses and their customers.
- A management dashboard providing real-time visibility into QR code usage, scan metrics, and tamper alerts.

4.1.3 Project Deliverables

Documentation

- Requirements and Analysis (R&A) Document
- System Architecture Design Document
- Project Management Plan including Scope, Schedule, and Risk Management
- Test Plan and Test Results Report
- User Manual
- Installation and Deployment Guide
- Final Project Report

Project Components

- Backend API server (Node.js + Express + PostgreSQL)
- Three web-based pages (Reacts):
- QR Code Landing Page — displayed when a user scans a QR code with a regular scanner (Apple/Google). Validates the token server-side and prompts the user to download the dedicated app for full security features.

- QR Code Display Page — shown after an admin generates a new QR code. Displays the generated QR code image and its details for printing or sharing.
- Admin Management Dashboard — a full web application for administrators to manage QR codes, review tamper alerts, and monitor scan metrics.
- Mobile application for QR code scanning and tamper alert reporting (React Native)
- PostgreSQL database with full schema for QR codes, scan logs, alerts, and users

4.1.4 Scope Statement

The Secure QR Code system addresses the inherent insecurity of publicly displayed QR codes, which can be physically replaced or tampered with by malicious actors. The system will deliver a full-stack authenticated QR code platform consisting of a backend API server, three web-based pages, and a mobile application.

What will be INCLUDED:

- QR code generation with cryptographically signed JWT tokens and configurable expiry.
- A server-side redirect and verification system that validates tokens before redirecting users to their intended destination.
- Backward compatibility with all standard QR code scanners — any device can scan the generated codes without special software.
- Three web pages serving distinct purposes:
- QR Code Landing Page: shown to regular scanner users when a QR code is scanned. Performs server-side token validation, displays a verification result, and prompts the user to download the dedicated app.

- QR Code Display Page: shown to admins after generating a new QR code. Displays the QR code image, destination URL, expiry date, and a download/print option.
- Admin Management Dashboard: a full web application for QR code management (generate, activate, blacklist), alert review with photo and GPS evidence, and scan metrics visualisation.
- User-agent-based detection to differentiate between dedicated app users and regular scanner users, delivering an appropriate experience to each.
- A blacklist and whitelist management system for flagging and deactivating compromised QR codes.
- A tamper alert system allowing mobile app users to report suspected tampering with GPS location, photo evidence, description, and optional contact details.
- Scan telemetry logging including timestamp, device type, IP address, and validation result.

What will NOT be included:

- Extension or modification of the ISO/IEC 18004 QR code standard (listed as an optional extra — outside current scope).
- Time-based redirect service (listed as an optional extra — outside current scope).
- Integration with third-party enterprise systems or external APIs beyond the core platform.
- User account registration or login for end-users scanning QR codes — authentication is token-based at the QR code level, not the individual user level.
- Native iOS app distribution via the Apple App Store (out of scope for this phase; Android-first development).
- Hardware or physical security of printed QR codes — the system cannot prevent physical replacement of codes, only detect and respond to it.

4.2 Work Breakdown Structure Summary

1. Project Management

1.1 Develop and maintain Project Management Plan

1.2 Conduct and document client meetings

1.3 Manage team task allocation and progress tracking

1.4 Conduct risk identification and mitigation planning

1.5 Prepare and submit milestone deliverables

2. Requirements and Analysis

2.1 Gather and document functional requirements from client brief

2.2 Identify and document non-functional requirements

2.3 Clarify open items with client (Peter and Mike)

2.4 Produce and submit Requirements and Analysis Document

2.5 Obtain client sign-off on requirements

3. System Architecture and Design

3.1 Define overall system architecture (backend, web pages, mobile app)

3.2 Design database schema (QR codes, scan logs, alerts, users)

3.3 Define API endpoint structure and contracts

3.4 Define token signing and validation approach (JWT)

3.5 Define user-agent detection logic for app vs browser users

3.6 Define the three web page flows and their purposes

3.7 Produce Architecture Design Document

4. Backend Development

4.1 Set up Node.js + Express project and PostgreSQL database

4.2 Implement database schema using Prisma ORM

4.3 Implement QR code generation endpoint (POST /api/qr/generate)

4.4 Implement token signing service using JWT

4.5 Implement QR verification and redirect endpoint (GET /api/qr/verify)

4.5.1 Token validation (signature, expiry)

4.5.2 Blacklist and whitelist check

4.5.3 User-agent detection and response branching

4.5.4 Scan logging to database

4.6 Implement alert submission endpoint (POST /api/alert/report)

4.6.1 Photo upload handling

4.6.2 GPS and description storage

4.6.3 Automatic QR code status update to suspicious

4.7 Implement admin API endpoints

4.7.1 Get all QR codes with scan and alert counts

4.7.2 Get individual QR code details with scan history

4.7.3 Update QR code status (blacklist / activate)

4.7.4 Get all tamper alerts

4.7.5 Get metrics and telemetry summary

5. Web Page Development

5.1 QR Code Landing Page (for regular scanner users)

5.1.1 Display token validation result (safe / warning / blacklisted)

5.1.2 Show destination URL and redirect option for valid codes

5.1.3 Display app download prompt and link

5.1.4 Display clear warning message for invalid or blacklisted codes

5.2 QR Code Display Page (for admins after generation)

5.2.1 Display generated QR code image

5.2.2 Show destination URL and expiry information

5.2.3 Provide download and print options for the QR code image

5.3 Admin Management Dashboard

5.3.1 Implement application layout and navigation sidebar

5.3.2 Implement Dashboard page with metrics cards and scan chart

5.3.3 Implement QR Code Management page

5.3.4 Implement Alerts review page with photo, GPS, and reporter details

5.3.5 Connect all pages to backend API

6. Mobile Application Development

6.1 Set up React Native project

6.2 Implement QR code scanner screen

6.3 Implement token submission to backend verify endpoint

6.4 Implement verification result screen (safe / warning)

6.5 Implement tamper alert reporting screen

6.5.1 GPS location capture

6.5.2 In-app camera for photo capture

6.5.3 Description and optional contact form

6.5.4 Alert submission to backend

6.6 Set custom User-Agent header (SecureQRApp/1.0) on all requests

7. Testing

7.1 Develop test plan covering all functional requirements

7.2 Unit testing of backend services (token, QR generation, alert)

7.3 Integration testing of all API endpoints

7.4 End-to-end testing of the full scan and alert flow

7.5 Backward compatibility testing with standard QR scanners

7.6 Web page testing across mobile and desktop browsers

7.7 User acceptance testing with client

7.8 Document and report test results

8. Documentation

8.1 Write and maintain Requirements and Analysis Document

8.2 Write Architecture Design Document

8.3 Write User Manual for admin dashboard and mobile app

8.4 Write Installation and Deployment Guide

8.5 Produce Final Project Report

5. Project Time Management (Felixia)

The work breakdown structure is developed for this project and included in Appendix B. It describes all activities necessary for finishing the project. About 65 separate activities are divided into eight activity categories: Project Management, Requirements and Analysis, System Architecture and Design, Backend Development, Web Page Development, Mobile Application Development, Testing, and Documentation.

All the activities are associated with durations measured in hours or days. Durations were determined by considering team members' experience in executing similar coding activities and difficulty levels of activities. Thus, the task to configure the backend project using Node.js and Express takes a day since team members have already accomplished this task during previous units. The activity to develop a complete admin dashboard including all web pages and API calls will take five days as there are many activities involved.

Dependencies exist between different tasks. In cases where Task B can only start after Task A is finished, a dependency exists. Some of the dependencies that have been identified in this QR code generation project include:

- QR Code landing page generation can only happen after the QR verification endpoint has been created.
- Mobile application scanning screen development can only happen after the QR verification endpoint has been created.
- Testing of tamper alert system can only begin after alert submission endpoint has been created.
- Compilation of user manuals can only be undertaken when the project is near completion because screens and buttons must be consistent with the final system.
- Critical path is that path along which all delays will affect the total project duration; therefore, delay in the execution of any task will affect the project deadline of 1 August 2026.
- The critical path in this project includes:
  Client signs off on the requirements → Design database schema → Token signing service → QR verification endpoint → Mobile scanner screen development → Alert submission endpoint → Testing of final system

If there is any delay with any of the above tasks, prompt action must be taken.

5.1 Project Milestone Dates

This assignment started on 20 May 2026. Every assignment will have to be done by the time indicated below. Deadline will be strictly observed for all deliveries. Any extension will only be granted if there is an emergency situation.

Deliverable	Due Date
Team Charter submission	23 May 2026
Project Management Plan V1.0	6 June 2026
Requirements and Analysis Document	8 June 2026
Design Document	27 June 2026
Self and Peer Evaluation 1 (SPE1)	27 June 2026
Final Project Submission (all code and all documents)	1 August 2026
Project Presentation PowerPoint file	1 August 2026
Demo media file	1 August 2026
Promo Video file	1 August 2026
Self and Peer Evaluation 2 (SPE2)	1 August 2026

5.2 Timeline for Development Work

Tasks for requirements finalization, database design and backend configuration will take place during the second sprint from June 6 to June 19, 2026. These tasks will form the basis for further development in the third sprint. It is worth noting that even with this change to the timeline, the requirement final submission date 1st August still remains realistic.

5.3 Gantt Chart

To help us track the entire project timelines, a Gantt chart has been made. In a Gantt chart, each task identified from the Work Breakdown Schedule appears on the Gantt chart in form of a horizontal bar on the calendar. The point where the bar starts denotes the starting date while the end of the bar denotes the ending of the task.

Dependencies can be easily detected through the drawing of lines linking the ending of one task with the beginning of another task.

The Gantt chart's timeline covers the period from 20th May 2026 to 1st August 2026.

The Gantt chart can be found in Appendix B of this document.

5.4 Network Diagram

A network diagram has been created to help better understand the relationships between tasks. In this network diagram, rectangles are used for tasks and arrows are used to indicate the task that should be completed before a new task begins.

From the network diagram, one realizes that the most crucial task in the whole project is the QR verification endpoint because it depends on five other tasks: landing page, mobile scanner, the process of redirection, the scan log capability, and the testing process.

The Network Diagram can be found in Appendix B of this document.

5.5 Weekly Progress Monitoring

Patricia, the project manager, will analyze the actual performance against the expected performance at 9:00AM every Monday morning based on the Gantt chart. Progress reporting will take place weekly via the communication channel of the MsTeams.

In case the project team falls behind schedule for more than two days, the conflict resolution procedure specified in Section 9 is employed. It includes the following steps:

1. The person in charge of the corresponding activity provides an explanation for the delay.

2. Team analyzes the problem from all angles and discusses it.

3. Team explores possible courses of action and agrees upon a corrective course of action. It involves either the additional work done by the person responsible for that activity or streamlining the activity by excluding unnecessary components or delaying another activity.

4. Outcome of the discussion is registered in the minutes.

The delays are always discussed and resolved immediately.

5.6 Buffer Days Prior to Deadlines

Since all deliverables are handed in strictly according to the set deadlines, buffer days have been added prior to each deadline. It means that all deliverables should be fully prepared two days prior to the deadline.

For instance, the deadline for the Final Project Submission is 1 August. As a result, the completion of all the code, writing all necessary documentation, and testing should take place by 30 July. Two days left will be the buffer period for addressing possible problems. In case any problem emerges in the course of work on 30 July, one can still fix them within the time period until 1 August.

Concerning the 1 August deadline, the period of five buffer days from 27 July until 31 July has been selected. This period of five days is intended solely for solving emergency problems that may emerge in the process of preparation of the deliverables. Any additional work is not expected within this timeframe.

5.7 Recording Actual Finish Dates Against Planned Finish Dates

An estimation comparison spreadsheet has been designed in Excel sheets where there are five columns namely Task ID, Task name, Planned finish date, Actual finish date, and days variance.

Whenever a particular task is completed by its assignee, the assignee will then enters the actual date in the relevant cell and the project manager calculates the variance. If the actual completion date and the planned completion date coincide, then the variance would be zero while in case of a two-day delay, variance would be +2.

Once two sprints are done, the variance column is verified for the precision of the estimate. If there is a significant amount of variance in the schedule, then the estimate and prioritization of tasks will be evaluated and modified if required.

The Gantt chart is reviewed and updated every Thursday after the weekly meeting of the team and the new version of it is named with the new date in it. The older versions will be kept in a separate folder.

6. Project Quality Management (Alvin and Vanessa)

Quality is described as the level to which a system fulfills the specified criteria and achieves its intended function. In this project:

- **Adherence to requirements:** Ensuring that the secure QR system fulfills the functional and security criteria, including QR, expiration verification and safe redirection.

- **Usability and effectiveness:** Guarantee that the system is functional, dependable and accessible for users on mobile.

6.1 Component quality checklist

Component	Checklist	
QR code generator	Creates unique UUID, scannable QR, free of duplicates	
QR validation API	Accurately checks QR and rejects expired or blacklisted codes.	
Mobile application	Functional QR scanning, QR verification and tamper report submission	
Redirect system	Redirects to the URL within ≤ 3 seconds	
Database	Data is stored without any corruptions with backups and recovery support.	
Reporting system	Accurately records GPS, description, photo evidence and submission timestamp.	
Admin dashboard	Correctly display the right alerts, system status information and scan metrics.	

6.2 Quality assurance testing

- Perform testing to check on the accessibility of the mobile application.

- Perform testing on the QR code with and without the mobile application to check if the redirect system is working.

- Perform testing to check if the download link for the mobile application is working.

- Perform testing to check if QR code has an expiry

- Conduct tests to ensure that the blacklisted QR codes are accurately identified and blocked from redirecting users.

- Conduct tests to verify whether alerts are correctly submitted and is visible in admin dashboard.

6.3 Performance metrics, integrity safeguards and acceptance criteria

QR code generator

Metrics:

- QR code generation success rate ≥ 99%

- Duplication rate of UUID = 0%

Acceptance criteria:

- Every single QR code generated is unique

- No duplicated QR codes in the database
- QR codes can be scanned successfully using the standard phone camera.

QR validation API:

Metrics:

- QR code expiry rate = 100%
- Blacklisted QR code detection rate = 100%

Acceptance criteria:

- QR code after the expiry period and invalid should be shown as invalid.
- Blacklisted QR codes must be blocked from redirecting the users.

Mobile application:

Metrics:

- Successfully QR code scan rate ≥ 99%
- Tamper report submission success rate ≥ 95%
- Crash/downtime rate ≤ 0.1%
- Average delay time < 3 seconds

Acceptance criteria:

- Users must be able to scan the QR code successfully
- Reports must be submitted successfully
- Alerts must be displayed correctly with accurate details (GPS, image, description, time)

<u>Redirect system:</u>

Metrics:

- Redirection of URL time ≤ 3 seconds

Acceptance criteria:

- Time taken for the redirection of URL to take place should be 3 seconds or less.

<u>Database:</u>

Metrics:

- Corrupted data in the database = 0%

Acceptance criteria:

- Any data stored in the database must be inherently uncorrupted

<u>Reporting system</u>

Metrics:

- Report submission rate = > 95%
- Accuracy of GPS rate > 95%
- Successful image upload rate > 95%

Acceptance criteria:

- All reports must be safely stored and can be retrieved at any point of time
- Reports must include accurate GPS location, description, timestamp and images

<u>Admin dashboard:</u>

Metrics:

- Accuracy of alerts shown to admin ≥ 99%

Acceptance criteria:

- Alert should contain description of device or location using mobile device information
- If the user wishes to share; reporter name and contact information
- Photo of QR code and surroundings

7. Project Communications Management (Qiyu)

Project communication management is about how we create and share information for the Secure QR Code Development project. We must store and update this information too. The Secure QR Code Development project has a lot of people involved like clients and supervisors and project managers and developers and testers and the people who do the documentation and business analysts. Thus, it is important that we all talk to each other in a way that makes sense.
The Secure QR Code Development project is about making a system to authenticate QR codes. This system includes making QR codes and a platform to manage them and a way to redirect people and a way to track how people use them and alerts if someone tries to tamper with them and a mobile app. The Secure QR Code Development project is complicated. We must make sure we communicate clearly and on time and, in a professional way and we must write everything down.

7.1 Stakeholder Analysis

The following stakeholders are involved in the project communication process:

Stakeholder	Role in Project	Communication Needs	Communication Method	Frequency
Peter Cole and Mike Groeneweg	Clients	Provide project expectations, confirm requirements, review milestones, provide feedback	Email, Microsoft Teams	After major milestones or when client clarification is required
Patricia Orenza	Project Manager and main contact person	Coordinate the team, assign work, track progress, communicate with clients and supervisor	WhatsApp, Microsoft Teams, Email, OneDrive	Daily or weekly as needed

Academic Supervisor	University project supervisor	Provide academic guidance, monitor team progress, clarify unit expectations	Microsoft Teams, Email, weekly meeting	Weekly or when required
Roy Yoo Hojin and Gu Qiyu	Technical Architect / Developers	Develop backend, dashboard, redirect system, database functions, and technical components	WhatsApp, GitHub, Microsoft Teams, OneDrive	Several times per week during development
Vanessa Tan	Tester	Develop backend, dashboard, redirect system, database functions, and technical components	WhatsApp, GitHub, OneDrive	During testing stages and before milestone submissions
Lim Yu Jun Felixia	Business Analyst	Record requirements, translate client needs into user stories, manage requirement changes	WhatsApp, OneDrive, Microsoft Teams	Weekly and during requirements updates
Alvin Rui Long Yeow	Documentation	Maintain meeting minutes and project documentation	OneDrive, WhatsApp	Weekly and before submissions

Whole Project Team	Project delivery team	Share updates, discuss problems, review tasks, complete deliverables	WhatsApp, Microsoft Teams, OneDrive, GitHub	Weekly meetings and ongoing messages

The main external stakeholders are Peter Cole and Mike Groeneweg, who are identified as the clients for the Secure QR Code Development project. The project objective is to build a secure QR code authentication and management system that supports expiry timeout, unique authentication codes, redirects, telemetry, alerts, and mobile scanning. The Team Charter also identifies Patricia Orenza as the Project Manager and main point of contact for the clients.

7.2 Communication Methods and Frequency

We will use different communication methods depending on the purpose and urgency of the information.

Team daily communication	**Quick questions, progress updates, reminders, issue discussion**	**WhatsApp**	**As needed, checked daily**	**All team members**
Weekly team meeting	Review progress, discuss problems, assign tasks, confirm next steps	Microsoft Teams or face-to-face meeting	Once per week, usually Thursday or Friday night	Project Manager
Client communication	Requirement clarification, milestone updates, approval, feedback	Email and Microsoft Teams	After major milestones or when needed	Patricia

Supervisor communication	Academic guidance, progress review, project advice	Microsoft Teams and Email	Weekly or when needed	Patricia and team
Document sharing	Store and share project documents, meeting minutes, plans, and reports	OneDrive	Updated whenever documents change	Documentation member and relevant task owner
Code collaboration	Store source code, track code changes, review development progress	GitHub	Updated during development	Developers
Testing communication	Report bugs, testing results, and fixes required	WhatsApp, GitHub Issues, OneDrive	During testing phase and before submissions	Tester and developers

7.3 Response Time and Communication Rules

To ensure that communication is effective and that project delays are avoided, the team will follow the response rules below:

Communication Situation	Expected Response Time
General WhatsApp message	Within 24 hours
Urgent project issue or deadline-related message	Within 6 hours
Client or supervisor email	Within 24–48 hours
Document review request	Within 2 working days

GitHub pull request or code review	Within 1–2 working days
Bug report during testing period	Within 24 hours

Every single team member needs to look at the communication channels we agreed on all the time. We should check WhatsApp every day because it is the way for the team to talk to each other quickly. Team members should check OneDrive before we have a meeting and before we must submit something. Developers and testers should check GitHub a lot when we are working on something and testing it.

If a team member is running late or cannot do a task on time they should tell the Project Manager away. This helps the team give the task to someone change the schedule or tell the supervisor, about the problem if we need to.

7.4 Tone of Communication

All project communication must be professional, respectful and clear. We need to be mindful of this because our project involves working with staff, clients and team members from different technical backgrounds. So, we will use a constructive tone in all our communication.

When we communicate within our team we can use direct language. We should still be respectful and focused on the task at hand. However, when we communicate with clients and supervisors we should use language.

Here are some guidelines for writing emails:

- Use a subject line
- Start with a greeting
- State the purpose of the message
- Specify the action
- Include a deadline if applicable
- End with a closing

Sometimes disagreements will happen. When they do, we will focus on the project issue not on criticism. We will discuss problems based on evidence project requirements and task priorities. If we can't resolve a disagreement, within our team the Project Manager will escalate the issue to the supervisor.

7.5 Documentation Formatting

We will write all project documents using Microsoft Word. These documents will be stored in a folder so the team can access them. We will use the format for all documents. This means all project work will look professional and be easy to understand. We will use Times Roman font for the main text. The font size will be 12 points. Section headings will have numbers like 8.1, 8.2 and 8.3. This will keep the project documents organised. Tables and images and charts and screenshots should be neat. They should be close to the text that explains what they mean. Team has not decided on a format for captions for charts and figures. We should explain important tables and charts in the text. This way readers can understand what they mean. Before we submit the documents, we should check them. We need to check for spelling mistakes and grammar mistakes and formatting and page numbers and version numbers. The final project submission will follow the guidelines, from the LMS and the assessment templates.

7.6 Technologies and Access Methods

The project will use the following technologies and access methods for communication, document management, and development:

Tool / Technology	**Purpose**	**Access Method**
WhatsApp	Fast team communication, reminders, informal updates	Team group chat
Microsoft Teams	Formal meetings, supervisor meetings, client meetings, online discussions	Team meeting links or invited channels
Email	Formal communication with clients, supervisor, and important approvals	University email accounts
OneDrive	Shared document storage, meeting minutes, reports, planning documents	Shared team folder
GitHub	Source code storage, version control, development collaboration, issue tracking	Shared GitHub repository
Microsoft Word / PDF	Project documentation and submission files	Stored in OneDrive
Excel or task board	Progress tracking and task management	Stored in OneDrive or shared workspace

We should only let the people who are supposed to be on our team use OneDrive and GitHub. We should only let clients look at things when they really need to like when we want them to check what we have done far or see how the project is going or look at what we have agreed to do. We must be careful with information like passwords API keys, database credentials and private tokens. We should never share OneDrive and GitHub secret information like passwords API keys, database credentials and private tokens on WhatsApp or, in documents that everyone can see.

7.7 Communication Plan Update Method

The Communication Management Plan will be reviewed regularly during the project to ensure that it remains suitable for the team’s working process. The plan will be reviewed at least once every two weeks or whenever a major communication issue occurs.

The plan may be updated when:

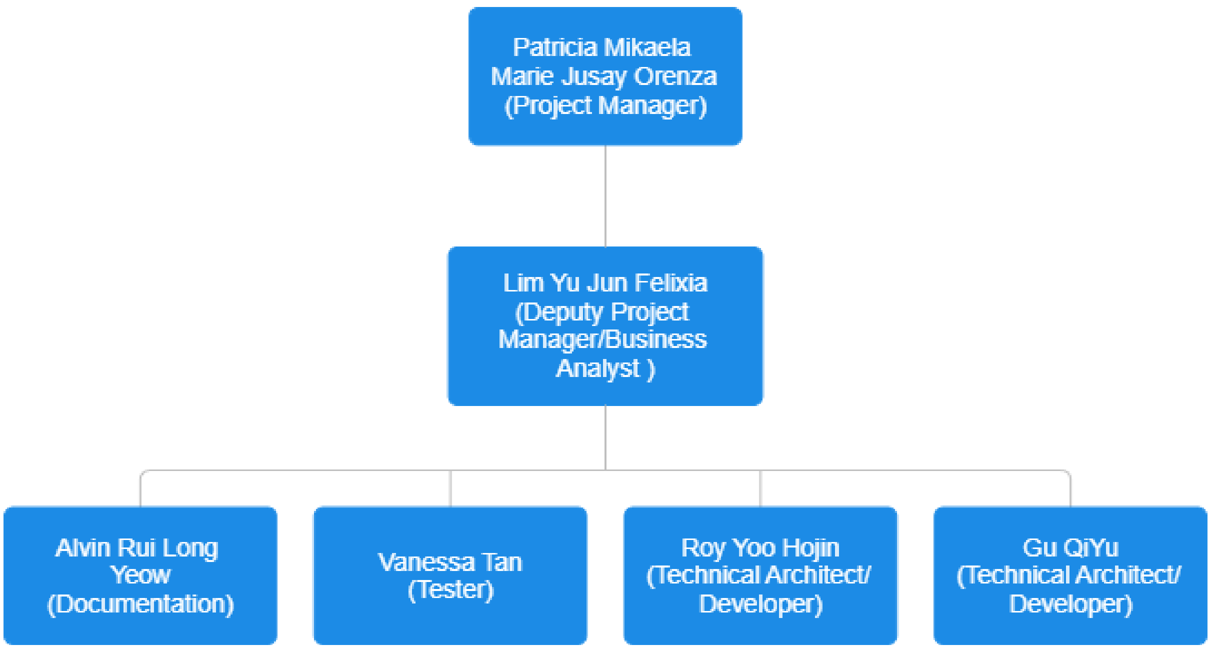
- a new communication tool is introduced;
- the meeting schedule changes;
- client or supervisor communication expectations change;
- team members experience delays due to unclear communication;
- document storage or version control methods need improvement;
- a communication-related risk is identified.

Any proposed update to the Communication Management Plan should first be discussed during a team meeting. If the change only affects internal team communication, the Project Manager may approve it after team agreement. If the change affects client communication, supervisor communication, deliverable submission, or official project records, Patricia will confirm the change with the supervisor or client before applying it.

All approved changes will be recorded in the meeting minutes and reflected in the latest version of the Project Management Plan stored in OneDrive.

8. Project Human Resources Management (Alvin)

Organisational Chart:



Responsibility Assignment Matrix:

		Resource					
	Tasks	Patricia	Felixia	Alvin	Vanessa	Roy	Qiyu
Project Management	Develop Project Management Plan	A/R	A	C	C	C	C
	Conduct client meetings	A/R	A/R	R	R	R	R
	First client meeting	A/R	A/R	R	R	R	R
	Second client meeting	A/R	A/R	A/R	A/R	A/R	A/R
	Client walk-through / UAT	A/R	A/R	A/R	A/R	A/R	A/R
	Task allocation and tracking	A/R	A	I	I	I	I
	Risk identification and mitigation	A	A	I	R	I	I
	Milestone deliverable preparation	A/R	A	C	C	C	C

Requirements and Analysis	Gather functional requirements	A	A/R				
	Gather non-functional requirements	A	A/R				
	Clarify open items with client	A	A/R	C	C	C	C
	QR code expiry duration	A	A/R			I	I
	User information fields	A	A/R			I	I
	Expected daily user volume	A	A/R			I	I
	Telemetry requirements	A	A/R			I	I
	Log retention period	A	A/R			I	I
	Produce R&A Document	A/R	A/R	R	R	R	R
	Obtain client sign-off	A/R	A	I	I	I	I
System Architecture and Design	Define overall system architecture					A/R	A/R
	Design database schema					A/R	A/R
	qr_codes table					A/R	A/R
	users table					A/R	A/R
	scan_logs table					A/R	A/R
	alerts table					A/R	A/R
	Define API endpoint structure					A/R	A/R
	Define token signing approach					A/R	A/R
	Define user-agent detection logic					A/R	A/R
	Define three web page flows					A/R	A/R
	Produce Architecture Design Document					A/R	A/R
Backend Development	Project and environment setup					A/R	A/R
	Database setup					A/R	A/R
	QR code generation endpoint					A/R	A/R
	JWT token signing					A/R	A/R
	QR image generation					A/R	A/R
	Database record creation					A/R	A/R
	Token service					A/R	A/R

Web Page Development	QR verification endpoint					A/R	A/R
	Token signature validation					A/R	A/R
	Token expiry check					A/R	A/R
	Blacklist and whitelist check					A/R	A/R
	User-agent detection and branching					A/R	A/R
	Scan logging					A/R	A/R
	Alert submission endpoint					A/R	A/R
	Photo upload handling					A/R	A/R
	GPS and form data storage					A/R	A/R
	Automatic status update					A/R	A/R
	Admin API endpoints					A/R	A/R
	GET /api/admin/qrcodes					A/R	A/R
	GET /api/admin/qrcodes/:id					A/R	A/R
	PATCH /api/admin/qrcodes/:id					A/R	A/R
	GET /api/admin/alerts					A/R	A/R
	GET /api/admin/metrics					A/R	A/R
	QR Code Landing Page					A/R	A/R
	Server-side token validation					A/R	A/R
	Safe result display					A/R	A/R
	Warning result display					A/R	A/R
	App download prompt					A/R	A/R
	Mobile responsive design					A/R	A/R
	QR Code Display Page					A/R	A/R
	QR code image display					A/R	A/R
	QR code details display					A/R	A/R
	Download option					A/R	A/R
	Print option					A/R	A/R
	Admin Management Dashboard					A/R	A/R
	Project and environment setup					A/R	A/R
	Application layout and navigation					A/R	A/R

	Dashboard metrics page					A/R	A/R
	QR Code Management page					A/R	A/R
	Generate QR code form					A/R	A/R
	QR codes listing table					A/R	A/R
	Blacklist and activate actions					A/R	A/R
	Alerts review page					A/R	A/R
	API integration					A/R	A/R
Mobile Application	Project and environment setup					A/R	A/R
	QR code scanner screen					A/R	A/R
	Token submission to backend					A/R	A/R
	Verification result screen					A/R	A/R
	Tamper alert reporting screen					A/R	A/R
	Automatic alert trigger					A/R	A/R
	Manual alert trigger					A/R	A/R
	GPS location capture					A/R	A/R
	In-app photo capture					A/R	A/R
	Description and contact form					A/R	A/R
	Alert submission to backend					A/R	A/R
	Custom User-Agent header					A/R	A/R
	App permissions configuration					A/R	A/R
Testing	Develop test plan			R	A/R		
	Backend unit testing			R	A/R	C	C
	Token generation tests			R	A/R	C	C
	Token validation tests			R	A/R	C	C
	QR generation tests			R	A/R	C	C
	API integration testing			R	A/R	C	C
	POST /api/qr/generate tests			R	A/R	C	C
	GET /api/qr/verify tests			R	A/R	C	C
	POST /api/alert/report tests			R	A/R	C	C
	Admin endpoint tests			R	A/R	C	C

	End-to-end flow testing			R	A/R	C	C
	Backward compatibility testing			R	A/R	C	C
	Web page testing			R	A/R	C	C
	Landing page tests			R	A/R	C	C
	QR display page tests			R	A/R	C	C
	Dashboard tests			R	A/R	C	C
	Mobile app testing			R	A/R	C	C
	User acceptance testing	A/R	A/R	I	I	I	I
	Document and report test results	A	R	R	A/R	C	C
	Requirements and Analysis Document		A/R	A/R			
	Architecture Design Document			A/R		C	C
	Project Management Plan	A/R	R	A/R	R	R	R
Documentation	User Manual			A/R		C	C
	Admin dashboard user manual			A/R		C	C
	Mobile app user manual			A/R		C	C
	Installation and Deployment Guide			A/R		C	C
	Final Project Report	A/R	A/R	A/R	A/R	A/R	A/R
Key:							

R: Responsible. They are people who put in the work to complete the tasks.
A: Accountable. People who are answerable for the task's completion by delegating the work and review to ensure it meets standards
C: Consulted. People who provide input based on their expertise
I: Informed. People who are informed or kept up to date on the progress of the task at hand

Our team will have weekly online meetings on Thursday or Friday nights to follow up on any tasks given and to keep track of each other's progress. Any other questions that we have can be asked and answered via our group chats or through this meeting.

To resolve any conflicts in the team, we will lay out the situation at hand and the perspectives of both sides of the situation. As a team, we will determine who is at fault and what cofmpromises both sides should make.

9. Project Risk Management (Vanessa)

Risk management involves the identification, evaluation, monitoring and response to occurrences that could negatively affect the project goals. The subsequent risk register outlines possible risks linked to the Secure QR Code Authentication System project, the probability and impact, the mitigation strategies and the members who are responsible, These risks will be constantly reviewed during our weekly meetings and changes will be made as necessary throughout the entire project duration.

9.1 Risk register

Risk ID	Risk name	Description	Risk category	Impact	Probability	Mitigation/ response strategy	Person responsible
R01	Scope creep	Extra requirements or requests of features that could potentially extend the project beyond the existing resources and time.	Project management	High	High	Prioritize the essential deliverables, uphold a clear project scope and categorize non-essential features as potential upgrades.	Patricia
R02	User refusing device permissions	Users might not allow permissions for the camera, storage or GPS which is required for QR scanning and the reporting of incidents.	Technical	High	Medium	Present explicit permission instructions within the application itself and offer manual reporting choices.	Roy, Qiyu
R03	Data leak or information exposure	Confidential information like location	Security	High	Low	Encryption of sensitive data stored in the	Vanessa, Alvin

		details, user personal information or incident logs could be exposed to someone with unauthorized access.				database, mandate safe communications using HTTPS and establish role-based access control.	
R04	QR Code replication attack	Authorized QR codes can be replicated and deployed in unauthorized places which misleads the users.	Security	High	Medium	Encourage and enable users to flag any suspicious QR code and implement anomaly identification based on scan location to detect any unusual geographic patterns and potential QR replication attacks.	Vanessa, Alvin
R05	Issues with mobile device compatibility	Applications might function differently depending on the OS versions or device models.	Technical	Medium	Low	Using various android versions and devices during the development process while conducting tests.	Roy, Qiyu
R06	Backend server downtime	Any server failures can hinder the operation of QR validation, authentication or redirection.	Infrastructure	High	Medium	Utilize reliable hosting with automated daily backups, implementing real-time monitoring including alerts that will trigger	Roy, Qiyu

						if downtime exceeds 3 minutes and ensuring to come up with a well-defined disaster recovery plan including service restart and database restoration.	
R07	Data loss or database malfunction	Records of QR, telemetry information and user submissions might be lost because of any corruption within the database or system malfunction.	Technical	High	Low	Set up automated daily backups of the database, database replication for extra security and establishing disaster recovery protocols to retrieve telemetry logs, QR data and user submissions during system downtime or database corruption.	Roy, Qiyu
R08	Concerns regarding APK sideloading trust	Users might be hesitant to install the application because of Android security alerts when activating “Install from Unknown Sources”	Business	High	Medium	Include precise installation guidelines with detailed steps with visual aids on the website to assist users in installation successfully.	Felixia, Patricia

		leading to decreased overall adoption of the system.					
R09	Incorrect QR classification	Valid QR codes might be mistakenly classified as harmful or suspicious.	Operational	High	Low	Set up validation guidelines, offer administrator evaluation processes and perform tests.	Vanessa, Alvin
R10	Low user adoption	Users might choose to use the standard built in phone camera scanners rather than downloading the secure QR application.	Business	High	Medium	Implement browser-based verification options and have clear and concise download instructions.	Felixia, Patricia
R11	Delay in project schedule	Development task might take longer than the expected duration causing a delay in the project milestones and final delivery.	Project management	High	High	Ensure to have regular progress checks, effectively allocate tasks and prioritise critical functionality.	Patricia
R12	SSL/TLS certificate expiry or misconfiguration (Security)	If the app depends on a particular certificate for communicating with the Node.js backend, if the certificate expires or is not updated	Security	High	Medium	Establish effective certificate lifecycle management and monitoring and guarantee that certificate renewal procedures are organized and	Roy, Qiyu (primary) + Vanessa, Alvin (security)

		properly, it can cause the application to not be able to connect to the server and cease functioning correctly.				communicated ahead of time to avoid any service interruptions.	
R13	GitHub merge conflicts	Work may be accidentally overwritten, or merge conflicts may occur due to concurrent code modification.	Technical	Medium	Medium	Utilize feature branches, code assessments, pull requests or regular commits.	Roy, Qiyu
R14	Malicious file upload	During submission of tampering report, attackers may include malicious files that are disguised as images.	Security	High	Medium	Restrict file types to approve only image formats, enforce size limits for files and validate MIME types.	Vanessa, Alvin
R15	Misunderstanding of requirement	Team may misunderstand what the requirements of the client are which can lead to the need to rework, changes to completed work or delays.	Requirements	High	Medium	Perform routine reviews, have regular showcase of progress and secure approval.	Patricia

R16	Unavailability of member	A member might become unavailable due to health issues, academic responsibilities or personal obligations.	Project management	High	Medium	Make sure that documentation is up to date, reassign task if needed, help each other out.	Patricia
R17	Increased system acceptance and stakeholder engagement	The Secure QR Code system experiences greater-than-anticipated adoption from both the businesses and users driven by significant interest in QR security resulting in enhanced feedback, utilization and potential scalability requirements that exceed initial projections.	Business / Opportunity	Medium	Medium	Design system architecture for scalability, confirm backend can accommodate higher demand and frequently monitor usage metrics. Make sure to be prepared for possible performance improvement or feature enhancement if usage surpasses expectations.	Patricia

9.2 Probability/impact matrix

IMPACT

PROBABILITY

	Low	Medium	High
High			R01, R11
Medium		R13, R17	R02,R04,R06,R10, R12,R14,R15,R16, R08
Low		R05	R09, R03,R07

Probability:
Low = Unlikely to happen
Medium = can happen throughout project duration
High = likely to happen
Impact:
Low = slight delay or no significant impact
Medium = influences module or sprint
High = threatens project completion or scope

10. Project Cost Management (Qiyu)

Project Cost Management explains how the project team will estimate, monitor and control any costs related to the Secure QR Code Development project. Since this is an academic project, we do not currently plan to purchase paid services or equipment. Most project activities will use existing student resources, free software tools, and university-supported platforms. The main cost of the project will be the time and effort contributed by team members, rather than direct financial expenses.

10.1 Cost Estimation

The Secure QR Code Development project is expected to have a low direct financial cost. Team members will use their own laptops for development, documentation, testing and communication. Microsoft Word, OneDrive, Microsoft Teams, Email, WhatsApp and GitHub will be used for project work, communication, document storage and source code management. These tools are either free, or already available to use, or supported through student accounts.

The project is expected to develop and test an Android version of the mobile application only. Therefore, no Apple Developer account or iOS App Store publishing cost is included in the current project scope. We also do not currently plan to purchase a domain name, SSL certificate, paid hosting service, paid database service, or paid app store deployment service. For database development, we may use local PostgreSQL or a free-tier cloud database service if needed. Any future paid service must be discussed with the Project Manager and approved before use.

10.2 Estimated Budget

Cost Item	Estimated Cost	Notes
Student laptops / personal devices	$0	Team members will use their own devices.
Microsoft Word, OneDrive, Teams and Email	$0	Available through existing student or university accounts.
WhatsApp	$0	Used for daily team communication.
GitHub	$0	Free version used for source code management and version control.
Local PostgreSQL database	$0	Used for development and testing if suitable.
Free-tier cloud database or hosting	$0	May be used only if needed and if free-tier options are available.
Android testing	$0	Testing will be done using Android devices or emulator where available.
Domain name / SSL certificate	$0	Not planned for the current stage.

App store publishing	$0	Play Store or App Store publishing is not planned for this phase.
Labour cost	$0	No paid labour cost because this is an academic project.

The estimated direct project budget is therefore $0 at the current planning stage.

10.3 Labour Cost

This project will not have any paid labour cost because the student team members will do all the work as part of the ICT302 unit. The time of each team member is still very important for this project. Each team member must finish the tasks they are given according to the project schedule. Roy and Gu Qiyu will do most of the development work for this project. Vanessa will help with the testing. The other team members will help with managing the project figuring out what is needed writing down what we do and talking to people, about the project.

Although labour is not financially charged, delays in task completion may still affect the project schedule and final deliverables. Therefore, the Project Manager will monitor progress through weekly meetings, task updates and milestone reviews.

10.4 Cost Control

To keep project costs down, we will not use paid tools or services unless they are really needed for the project. We will use tools or tools we already have whenever we can. If we need to use a paid service on the team member who wants to use it must tell us why we need it how much it will cost what benefits it will bring and if there are any free alternatives we can use instead. The Project Manager will talk to the team about the request. Get approval from the client or academic supervisor if we need to.

We will also try to reduce the risk of spending much money by only testing the project, on Android using free repositories on GitHub storing documents on OneDrive and using databases that are free or do not cost much. This way we can make sure the project is something we can do within the academic project environment.

11. Project Procurement Management (Felixia)

The concept of project procurement involves the acquisition of any external goods and services required by the project from an outside source. In our project called Secure QR Code Development, all resources that may be required can already be accessed by us without spending money. We don't plan on purchasing any paid software, paid cloud services, or paid hardware. Nevertheless, we have to record those resources which we use and how we could access them in case of a need.

Requirements for the Hardware and Software in this Project

Below are listed those hardware and software requirements for the completion of the project. Every team member will use his/her own laptop/computer for coding, writing reports, conducting tests, etc. There is no specific hardware required to purchase.

The software used by the group includes:

- Microsoft Word – for developing all project documents. It can be obtained from the university student accounts.
- Microsoft Teams and Email – for official correspondence between the team members and client/supervisor. Both programs can be obtained from the university student accounts.
- WhatsApp – for everyday communication between the team members. It is a free application for smartphones.
- OneDrive – for storage and distribution of project documents, minutes of meetings, and reports. They can be accessed through university student accounts.
- GitHub – for storing the code, version control, and task management. Free version is enough for this project.
- Microsoft Project – for creating Gantt chart and project schedule. It can be downloaded from university lab computers/student accounts.
- Microsoft Visio – for developing the network diagram. It can be downloaded from university lab computers/student accounts.

- Node.js, Express, PostgreSQL, React, React Native – for developing the system itself. All the above programs are completely free.
- Postman – for API testing. Free version is enough.

No cloud server hosting, paid databases, domain names, and other similar items will be purchased for this project. Everything will be developed on our computers.

Product Assessment

The Technical Architect and developers, Roy Yoo Hojin and Gu Qiyu, shall be tasked with assessing the technical products or tools required for development. They will first determine whether there is an option available that would allow the product to be acquired free of charge before recommending that the organization pay for anything. In addition, Felixia, the Business Analyst, will evaluate the requirements and documentation management tools. The project management tools shall be assessed by Patricia, the Project Manager.

In view of the fact that this is an academic exercise without funding for any paid services, our team will have to undertake the following process prior to using any paid product.

1. Team member seeking to use the product will search for its free alternative. There are many free products out there for development, testing, documentation, and communications purposes.
2. Where there are no free alternatives, the team member will submit a brief justification highlighting reasons for requiring the product; its cost, required period, and inability of free alternatives to meet the purpose.
3. Project Manager will propose the request in the upcoming team meeting. The project manager, along with some other members of the team, will examine the request and find out if it really is necessary to use the product.
4. Where team decides in favor of using the product, the Project Manager will seek supervisor's permission for purchase.
5. After approval from the supervisor, the team will ascertain if the product is really essential for project completion.

Where there are free products, procurement process is not necessary. It would be enough for a team member to download and utilize the product, however, he or she should make the other members know of the product through WhatsApp message within 24 hours.

When it comes to any product that saves data or source code, we have to be sure it does not contradict our current tools in any way. For instance, where we add a document saving product, it should function well alongside OneDrive. In case of a new code saving product, it should work together with GitHub.

The list of tools that are used during the course of the project will be maintained by the Project Manager, who will put it into the OneDrive account and keep updating it when the project adopts new tools. This project does not require any paid tool or service.

12. Conclusion (Felixia)

The Project Management Plan will serve as the primary means of planning and managing the delivery of the Secure QR Code Authentication System to be delivered to the clients, namely, Peter Cole and Mike Groeneweg. The reason why this document was prepared is to integrate all aspects of project management in one coherent plan that all members of the team can adhere to.

The document provided insight into the ten key areas of project management needed in the development of this project. Project Integration Management gave information about teamwork, change management, and the rationale behind two-week sprint cycle. The scope of the project was provided by the Project Scope Management section. The methodology for creating the time management plan was provided by the Project Time Management. The Project Quality Management provided quality assurance measures and indicators used to ensure that all components function as expected. How the team and stakeholder communicated among themselves was detailed in Project Communications Management. Roles and responsibilities of different team members and conflict resolution strategies were indicated in Project Human Resources Management. Project risk management identified 17 potential risks associated with this project. Project cost management assured us that there is no budget attached to this project. Project procurement management detailed the tools used in developing the project and acquiring paid products.

A complete copy of the Work Breakdown Structure, the Gantt chart, the network diagram, Team Charter, glossary of terms, and agenda and minutes of meetings with the client and supervisor are available in the appendices attached at the back of this document.

This document will keep on changing as time progresses, and team members will review it periodically during the entire project. Any changes that occur in the plan will have to go through the change management process discussed in section four. The deliverables for the entire project are expected to be delivered as per the timeline described in the Project Management Plan, with the last deliverable due by 1st August 2026.

13. Appendices (All)

13.1 Appendix A: Deliverable Task Breakdown Statement

Deliverable	Key Tasks	Responsible Member	Acceptance Criteria
Team Charter	Define roles, communication rules, project objectives	All Team Members	Charter approved and submitted
Project Management Plan	Scope, schedule, risk, quality, communications planning	Patricia	PMP submitted by due date
Requirements & Analysis Document	Gather and document functional and non-functional requirements	Felixia	Requirements approved by client and supervisor
Design Document	Architecture design, database design, API design	Roy, Qiyu	Design document completed and submitted
Backend Development	QR generation, verification, alerts APIs	Roy, Qiyu	Backend functions correctly and passes testing
Web Development	Landing page, QR display page, admin dashboard	Roy, Qiyu	Pages function according to requirements
Mobile Application	QR scanner, verification, tamper reporting	Roy, Qiyu	Application functions correctly on Android devices

Testing	Unit, integration, system and user acceptance testing	Vanessa	Test report completed
Documentation	User manual, deployment guide, final report	Alvin, Felixia	Documentation completed and submitted
Final Submission	Compile all deliverables and submit	All Team Members	All project deliverables submitted by deadline

13.2 Appendix B: Work Breakdown Structure and Schedule

WBS (Work Breakdown Structure)

4.2.1 Project Management

ID	Task Name	Description	Owner	Deliverable
1	**Project Management**	**All activities relating to managing the project lifecycle**	**PM**	**PMP**
1.1	Develop Project Management Plan	Create the initial PMP covering scope, schedule, risk, and communication	PM	PMP
1.2	Conduct client meetings	Organise, facilitate, and document all meetings with Peter and Mike	PM	Meeting Minutes
1.2.1	First client meeting	Gather initial requirements and clarify the project brief	PM	Meeting Minutes
1.2.2	Second client meeting	Confirm requirements, clarify open items, present architecture	PM	Meeting Minutes
1.2.3	Client walk-through / UAT	Present completed system to client for formal acceptance	PM	UAT Sign-off
1.3	Task allocation and tracking	Assign WBS tasks to team members and track progress throughout the project	PM	Task Board
1.4	Risk identification and mitigation	Identify project risks and define mitigation strategies in the risk register	PM	Risk Register

1.5	Milestone deliverable preparation	Prepare and submit each milestone deliverable on schedule	PM	All Deliverables

4.2.2 Requirements and Analysis

ID	Task Name	Description	Owner	Deliverable
2	**Requirements and Analysis**	**All activities relating to gathering, analysing, and documenting requirements**	**BA**	**R&A Document**
2.1	Gather functional requirements	Extract and document functional requirements from the client brief and meetings	BA	R&A Document
2.2	Gather non-functional requirements	Identify performance, security, compatibility, and scalability requirements	BA	R&A Document
2.3	Clarify open items with client	Follow up with Peter and Mike on unanswered questions from the brief	BA	Meeting Minutes
2.3.1	QR code expiry duration	Confirm acceptable expiry timeframes for generated QR codes	BA	Meeting Minutes
2.3.2	User information fields	Confirm what user data fields are required per QR code	BA	Meeting Minutes
2.3.3	Expected daily user volume	Confirm anticipated scan volume to inform capacity planning decisions	BA	Meeting Minutes

2.3.4	Telemetry requirements	Confirm what metrics and data the client expects the system to track	BA	Meeting Minutes
2.3.5	Log retention period	Confirm how long scan logs and alert records should be retained	BA	Meeting Minutes
2.4	Produce R&A Document	Write the full Requirements and Analysis document for client review and sign-off	BA	R&A Document
2.5	Obtain client sign-off	Have Peter and Mike formally approve the R&A document before development begins	PM	Signed R&A

4.2.3 System Architecture and Design

ID	Task Name	Description	Owner	Deliverable
3	**System Architecture and Design**	**All design activities for the overall system structure and technical decisions**	**DEV**	**Architecture Document**
3.1	Define overall system architecture	Design the layered system: backend API, three web pages, and mobile app	DEV	Architecture Document
3.2	Design database schema	Define all tables and their fields in PostgreSQL via Prisma	DEV	Schema Diagram
3.2.1	qr_codes table	Fields: id, token, destinationUrl, status, expiresAt, createdAt	DEV	Schema Diagram

3.2.2	users table	Fields: id, email, role, organisationId, createdAt	DEV	Schema Diagram
3.2.3	scan_logs table	Fields: id, qrCodeId, scannedAt, ipAddress, userAgent, gpsLat, gpsLng, result	DEV	Schema Diagram
3.2.4	alerts table	Fields: id, qrCodeId, reporterName, contactInfo, gpsLat, gpsLng, photoUrl, description, createdAt	DEV	Schema Diagram
3.3	Define API endpoint structure	Document all API routes, HTTP methods, request inputs, and expected responses	DEV	Architecture Document
3.4	Define token signing approach	Specify JWT algorithm, payload structure (qrCodeId, destinationUrl, expiry), and secret key management	DEV	Architecture Document
3.5	Define user-agent detection logic	Specify how app users (SecureQRApp/1.0) are differentiated from browser users	DEV	Architecture Document
3.6	Define three web page flows	Document purpose, inputs, outputs, and user journey for each of the three web pages	DEV	Architecture Document
3.7	Produce Architecture Design Document	Write full architecture document including diagrams, schema, and design rationale	DEV	Architecture Document

4.2.4 Backend Development

ID	Task Name	Description	Owner	Deliverable
4	**Backend Development**	**All server-side development using Node.js, Express, and PostgreSQL**	**DEV**	**Backend Server**
4.1	Project and environment setup	Initialise Node.js + Express project, install all dependencies, configure .env file	DEV	Backend Server
4.2	Database setup	Set up PostgreSQL, configure Prisma ORM, and run the initial schema migration	DEV	Database
4.3	QR code generation endpoint	Implement POST /api/qr/generate to create signed, expiring QR codes	DEV	Backend Server
4.3.1	JWT token signing	Sign a JWT token containing qrCodeId, destinationUrl, and expiry timestamp	DEV	Token Service
4.3.2	QR image generation	Generate a base64 PNG QR code image encoding the full verify URL	DEV	Backend Server
4.3.3	Database record creation	Save the new QR code record to the database with status 'active' and expiry timestamp	DEV	Database

4.4	Token service	Implement generateToken() and verifyToken() reusable utility functions	DEV	Token Service
4.5	QR verification endpoint	Implement GET /api/qr/verify to validate tokens and route users appropriately	DEV	Backend Server
4.5.1	Token signature validation	Verify JWT signature — reject any token that has been tampered with	DEV	Backend Server
4.5.2	Token expiry check	Reject tokens where the expiry timestamp has passed	DEV	Backend Server
4.5.3	Blacklist and whitelist check	Block QR codes with blacklisted status; allow active and whitelisted codes	DEV	Backend Server
4.5.4	User-agent detection and branching	App users (SecureQRApp) get direct redirect; browser users receive the landing page HTML	DEV	Backend Server
4.5.5	Scan logging	Log every scan attempt to scan_logs with device info, GPS (if provided), and result	DEV	Database
4.6	Alert submission endpoint	Implement POST /api/alert/report to	DEV	Backend Server

		receive and store tamper reports		
4.6.1	Photo upload handling	Accept image file uploads and store using multer; generate accessible photo URL	DEV	Backend Server
4.6.2	GPS and form data storage	Save GPS coordinates, description, and optional reporter contact info to alerts table	DEV	Database
4.6.3	Automatic status update	Update the flagged QR code's status to 'suspicious' on alert submission	DEV	Database
4.7	Admin API endpoints	Implement all API endpoints consumed by the admin management dashboard	DEV	Backend Server
4.7.1	GET /api/admin/qrcodes	Return all QR codes with total scan count and alert count per code	DEV	Backend Server
4.7.2	GET /api/admin/qrcodes/:id	Return a single QR code with its full scan history and associated alerts	DEV	Backend Server
4.7.3	PATCH /api/admin/qrcodes/:id	Update a QR code's status — accepted values: active, blacklisted, suspicious, expired	DEV	Backend Server

4.7.4	GET /api/admin/alerts	Return all tamper alerts with associated QR code destination and status	DEV	Backend Server
4.7.5	GET /api/admin/metrics	Return aggregated statistics: QR code counts by status, scan totals, and alert totals	DEV	Backend Server

4.2.5 Web Page Development

ID	Task Name	Description	Owner	Deliverable
5	**Web Page Development**	**Development of the three web-based pages that form part of the system**	**DEV**	**Web Pages**
5.1	QR Code Landing Page	Shown to regular scanner users (Apple/Google) after scanning a QR code	DEV	Landing Page
5.1.1	Server-side token validation	Validate the token from the URL query string against the backend before rendering	DEV	Landing Page
5.1.2	Safe result display	Show green verified page with 'Continue to destination' button for valid tokens	DEV	Landing Page
5.1.3	Warning result display	Show red warning page for expired, invalid, or blacklisted tokens	DEV	Landing Page

5.1.4	App download prompt	Display download link and message encouraging installation of the dedicated mobile app	DEV	Landing Page
5.1.5	Mobile responsive design	Ensure the page renders correctly across all mobile screen sizes and browsers	DEV	Landing Page
5.2	QR Code Display Page	Shown to admins after a new QR code is successfully generated	DEV	Display Page
5.2.1	QR code image display	Render the generated QR code image prominently so it can be scanned immediately	DEV	Display Page
5.2.2	QR code details display	Show destination URL, QR code ID, creation date, and expiry date	DEV	Display Page
5.2.3	Download option	Allow admin to download the QR code image as a PNG file to their device	DEV	Display Page
5.2.4	Print option	Allow admin to send the QR code image directly to a printer from the browser	DEV	Display Page
5.3	Admin Management Dashboard	Full React web application for system administrators to manage the entire platform	DEV	Dashboard

5.3.1	Project and environment setup	Initialise React project, install dependencies: axios, react-router-dom, recharts	DEV	Dashboard
5.3.2	Application layout and navigation	Build sidebar navigation linking Dashboard, QR Codes, and Alerts pages	DEV	Dashboard
5.3.3	Dashboard metrics page	Stat cards (total, active, suspicious, blacklisted, alerts) and scan overview bar chart	DEV	Dashboard
5.3.4	QR Code Management page	Generate QR codes, view all codes in table, activate or blacklist individual codes	DEV	Dashboard
5.3.4.1	Generate QR code form	URL input + expiry hours input; displays QR image and expiry info on success	DEV	Dashboard
5.3.4.2	QR codes listing table	Table showing all QR codes with status badge, scan count, alert count, expiry, and actions	DEV	Dashboard
5.3.4.3	Blacklist and activate actions	Buttons to update individual QR code status directly from the management table	DEV	Dashboard
5.3.5	Alerts review page	List all tamper alerts showing photo, GPS coordinates, description, and reporter details	DEV	Dashboard

5.3.6	API integration	Connect all dashboard pages to their respective backend API endpoints using axios	DEV	Dashboard

4.2.6 Mobile Application Development

ID	Task Name	Description	Owner	Deliverable
6	**Mobile Application**	**React Native mobile app for QR scanning, verification, and tamper alert reporting**	**DEV**	**Mobile App**
6.1	Project and environment setup	Initialise React Native project, install dependencies, configure Android/iOS permissions	DEV	Mobile App
6.2	QR code scanner screen	Implement camera-based QR code scanning using react-native-camera or Expo Camera	DEV	Mobile App
6.3	Token submission to backend	On scan, extract the token from the URL and send GET request to /api/qr/verify	DEV	Mobile App
6.4	Verification result screen	Display a green (safe) or red (warning) result screen based on the server response	DEV	Mobile App
6.5	Tamper alert reporting screen	Full in-app flow for reporting suspected QR code tampering	DEV	Mobile App
6.5.1	Automatic alert trigger	Auto-display the alert prompt when server	DEV	Mobile App

		returns suspicious or blacklisted status		
6.5.2	Manual alert trigger	Provide a 'Report Tampering' button accessible at any point after scanning	DEV	Mobile App
6.5.3	GPS location capture	Request location permission and capture precise GPS coordinates at the moment of reporting	DEV	Mobile App
6.5.4	In-app photo capture	Open device camera within the app for user to photograph the QR code and surroundings	DEV	Mobile App
6.5.5	Description and contact form	Text input for description; optional fields for reporter name and contact information	DEV	Mobile App
6.5.6	Alert submission to backend	Submit GPS, photo, description, and optional contact to POST /api/alert/report	DEV	Mobile App
6.6	Custom User-Agent header	Set 'SecureQRApp/1.0' as User-Agent on all HTTP requests so server identifies app users	DEV	Mobile App
6.7	App permissions configuration	Configure location, camera, and internet permissions in Android manifest and iOS plist	DEV	Mobile App

4.2.7 Testing

ID	Task Name	Description	Owner	Deliverable
7	**Testing**	**All testing activities to verify the system meets its functional and non-functional requirements**	**TST**	**Test Report**
7.1	Develop test plan	Write a test plan covering all functional requirements, test cases, and acceptance criteria	TST	Test Plan
7.2	Backend unit testing	Test individual backend service functions in isolation	TST	Test Report
7.2.1	Token generation tests	Verify tokens are correctly signed and contain the expected payload fields	TST	Test Report
7.2.2	Token validation tests	Verify expired, tampered, and valid tokens return correct validation results	TST	Test Report
7.2.3	QR generation tests	Verify generated QR codes encode a valid, browser-scannable HTTPS URL	TST	Test Report
7.3	API integration testing	Test all API endpoints using Postman with valid, invalid, and edge-case inputs	TST	Test Report
7.3.1	POST /api/qr/generate tests	Test with valid URL, missing URL, missing expiry, and boundary expiry values	TST	Test Report

7.3.2	GET /api/qr/verify tests	Test valid token, expired, tampered, blacklisted, and app vs browser user-agent scenarios	TST	Test Report
7.3.3	POST /api/alert/report tests	Test with full data, missing required fields, and without an attached photo	TST	Test Report
7.3.4	Admin endpoint tests	Test all admin routes for correct data retrieval, filtering, and status update behaviour	TST	Test Report
7.4	End-to-end flow testing	Test the complete user journey: generate QR → scan → verify → redirect or alert	TST	Test Report
7.5	Backward compatibility testing	Confirm QR codes are successfully scannable by Apple and Google built-in scanners	TST	Test Report
7.6	Web page testing	Test all three web pages across mobile and desktop browsers	TST	Test Report
7.6.1	Landing page tests	Test correct display for valid, expired, invalid, and blacklisted token scenarios	TST	Test Report
7.6.2	QR display page tests	Verify QR image renders correctly and download/print functions operate as expected	TST	Test Report

7.6.3	Dashboard tests	Verify all data loads correctly and admin actions (blacklist, activate) function properly	TST	Test Report
7.7	Mobile app testing	Test QR scanning, verification result display, and full alert submission on a physical device	TST	Test Report
7.8	User acceptance testing	Present the completed system to Peter and Mike for formal acceptance testing and sign-off	PM	UAT Sign-off
7.9	Document and report test results	Record all test cases, pass/fail results, defects found, and resolution status	TST	Test Report

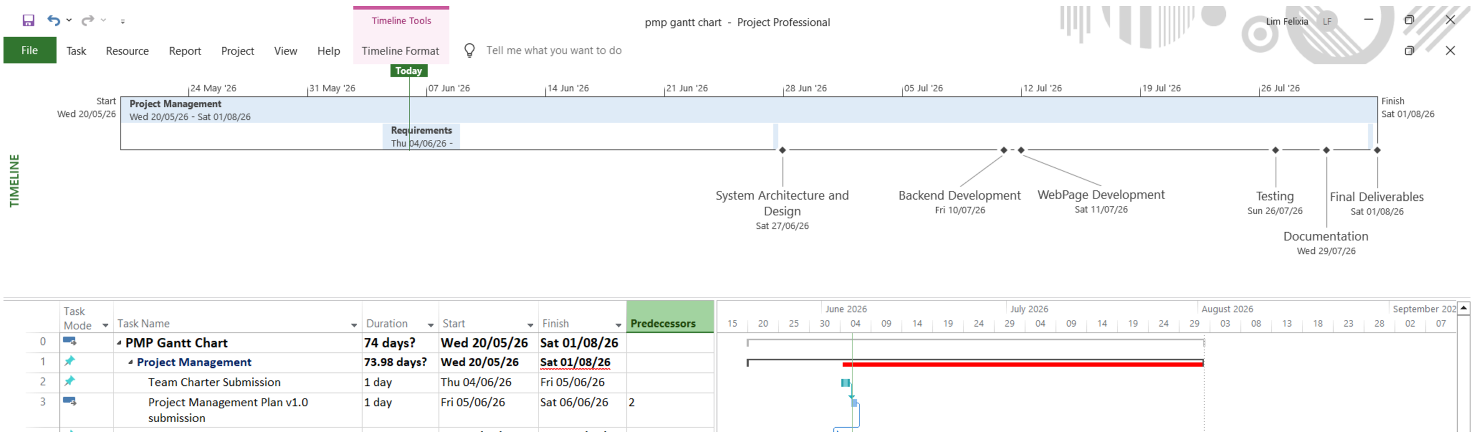
4.2.8 Documentation

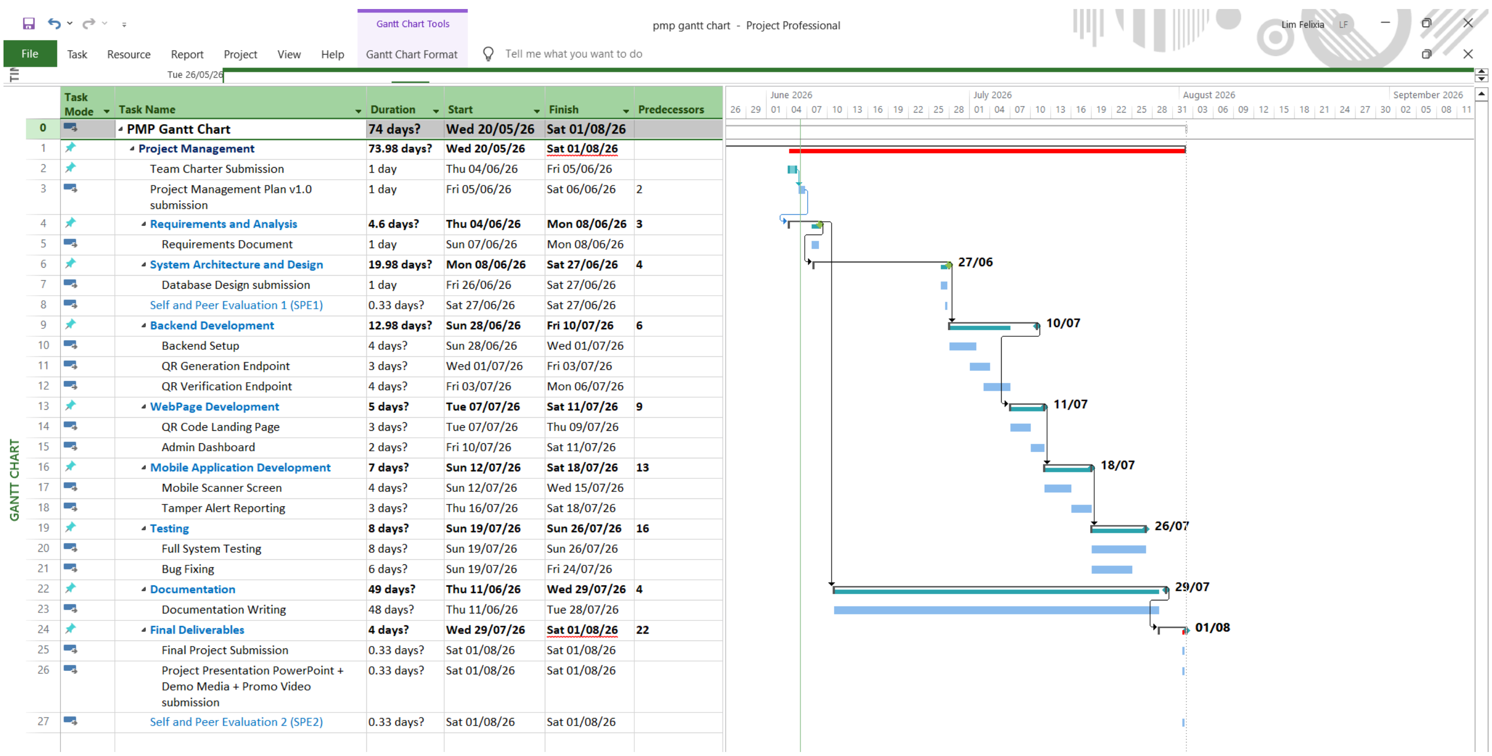
ID	Task Name	Description	Owner	Deliverable
8	**Documentation**	**All project documentation deliverables required for submission and client handover**	**DOC**	**All Documents**
8.1	Requirements and Analysis Document	Full R&A document covering all functional and non-functional requirements	BA / DOC	R&A Document
8.2	Architecture Design Document	System diagrams, database schema, API contracts, tech stack	DEV / DOC	Architecture Document

		rationale, and design decisions		
8.3	Project Management Plan	PMP covering scope statement, WBS, Gantt chart, risk register, and communication plan	PM / DOC	PMP
8.4	User Manual	Step-by-step usage guide for the admin dashboard and mobile application	DOC	User Manual
8.4.1	Admin dashboard user manual	How to log in, generate QR codes, manage the blacklist, and review tamper alerts	DOC	User Manual
8.4.2	Mobile app user manual	How to scan QR codes, interpret results, and submit a tamper report	DOC	User Manual
8.5	Installation and Deployment Guide	Step-by-step instructions for deploying the backend, dashboard, and mobile app	DEV / DOC	Deployment Guide
8.6	Final Project Report	Comprehensive report covering the entire project lifecycle, outcomes, and team reflections	ALL	Final Report

Key:
PM: Project Manager
BA: Business Analyst
DEV: Developer
TST: Tester
DOC: Documentation

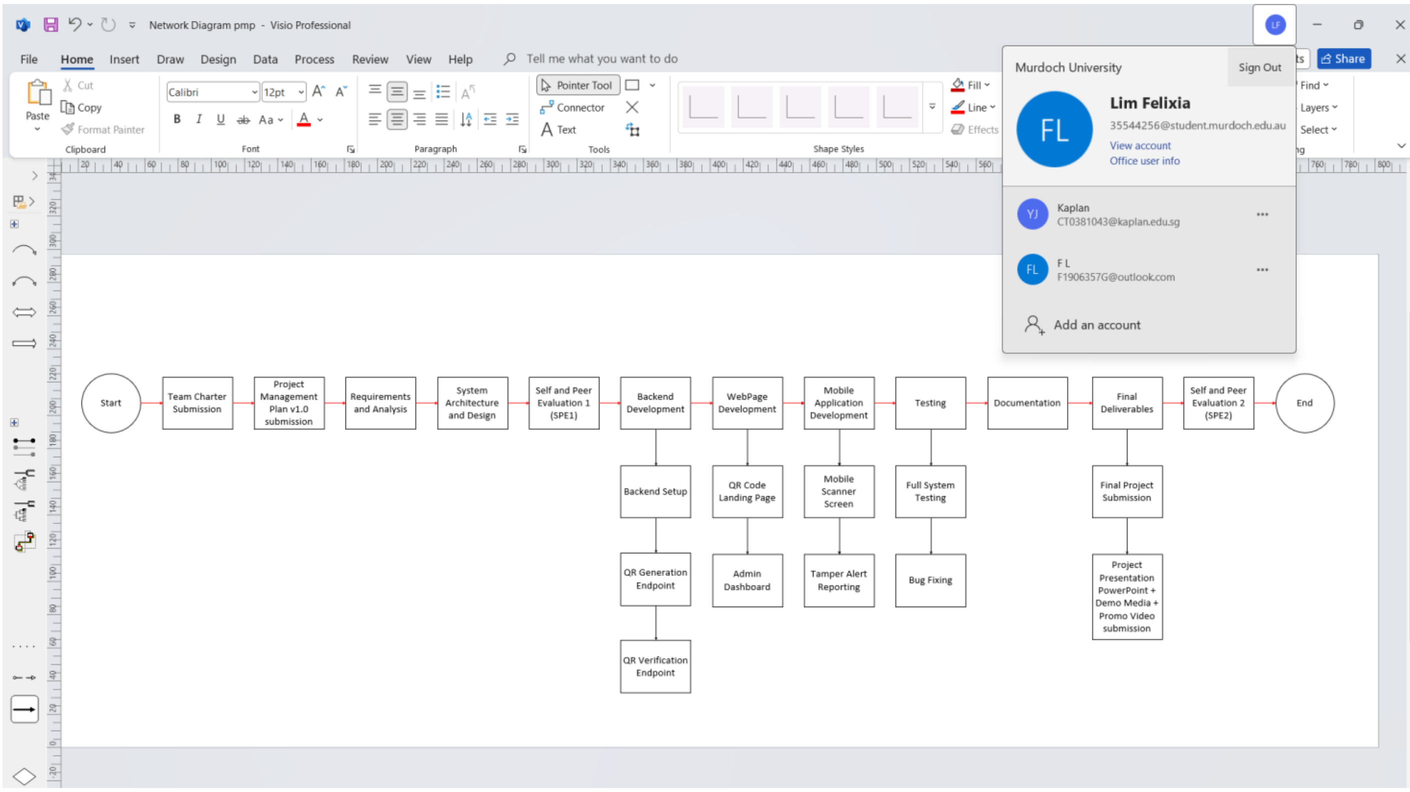
5.3 Gantt Chart





	Task Name	Duration	Start	Finish	Predecessors
0	PMP Gantt Chart	74 days?	Wed 20/05/26	Sat 01/08/26	
1	Project Management	73.98 days?	Wed 20/05/26	Sat 01/08/26	
2	Team Charter Submission	1 day	Thu 04/06/26	Fri 05/06/26	
3	Project Management Plan v1.0 submission	1 day	Fri 05/06/26	Sat 06/06/26	2
4	Requirements and Analysis	4.6 days?	Thu 04/06/26	Mon 08/06/26	3
5	Requirements Document	1 day	Sun 07/06/26	Mon 08/06/26	
6	System Architecture and Design	19.98 days?	Mon 08/06/26	Sat 27/06/26	4
7	Database Design submission	1 day	Fri 26/06/26	Sat 27/06/26	
8	Self and Peer Evaluation 1 (SPE1)	0.33 days?	Sat 27/06/26	Sat 27/06/26	
9	Backend Development	12.98 days?	Sun 28/06/26	Fri 10/07/26	6
10	Backend Setup	4 days?	Sun 28/06/26	Wed 01/07/26	
11	QR Generation Endpoint	3 days?	Wed 01/07/26	Fri 03/07/26	
12	QR Verification Endpoint	4 days?	Fri 03/07/26	Mon 06/07/26	
13	WebPage Development	5 days?	Tue 07/07/26	Sat 11/07/26	9
14	QR Code Landing Page	3 days?	Tue 07/07/26	Thu 09/07/26	
15	Admin Dashboard	2 days?	Fri 10/07/26	Sat 11/07/26	
16	Mobile Application Development	7 days?	Sun 12/07/26	Sat 18/07/26	13
17	Mobile Scanner Screen	4 days?	Sun 12/07/26	Wed 15/07/26	
18	Tamper Alert Reporting	3 days?	Thu 16/07/26	Sat 18/07/26	
19	Testing	8 days?	Sun 19/07/26	Sun 26/07/26	16
20	Full System Testing	8 days?	Sun 19/07/26	Sun 26/07/26	
21	Bug Fixing	6 days?	Sun 19/07/26	Fri 24/07/26	
22	Documentation	49 days?	Thu 11/06/26	Wed 29/07/26	4
23	Documentation Writing	48 days?	Thu 11/06/26	Tue 28/07/26	
24	Final Deliverables	4 days?	Wed 29/07/26	Sat 01/08/26	22
25	Final Project Submission	0.33 days?	Sat 01/08/26	Sat 01/08/26	
26	Project Presentation PowerPoint + Demo Media + Promo Video submission	0.33 days?	Sat 01/08/26	Sat 01/08/26	
27	Self and Peer Evaluation 2 (SPE2)	0.33 days?	Sat 01/08/26	Sat 01/08/26	

5.4 Network Diagram



13.3 Appendix C – Project/Team charter

Project Objectives:

To create and build a secure authentication and management system for QR codes that:

- Stops and identifies tampering with publicly accessible QR codes.
- Delivers a management system featuring expiration, distinct authorization codes, redirects, and tracking.
- Features a mobile application for scanning and verifying QR codes. Stays compatible with regular QR scanners (iOS/Android native).
- Allows users to report disruptions related to location, images, and optional contact details.

Approach: Summary of the stakeholder's needs and expectations, important assumptions, constraints. Refer to Project Management Plan and Communication Management Plan.

- We will meet every week to check our progress and plan the next steps.
- Each meeting will last around 2 to 3 hours.
- Between meetings, we will work on our own tasks and message each other on WhatsApp if we have questions.

Stakeholders: Peter Cole and Mike Groeneweg (client)

Assumptions:

- Users have smartphones with cameras and GPS.
- QR codes only for verified users and only able to access through the mobile application me made.
-

Constraints:

- Time: Complete by the end of TMA2026
- Must work on android
- Skills: not everyone knows how to do mobile development need to learn new skills and research

Roles and Responsibilities:

Role	Name	Position/ Key Responsibilities
Project Manager	Patricia Mikaela Marie Jusay Orenza	• Create and maintain the project schedule with clear deadlines. • Break down, assign work so that everyone has equal work

		• Track progress of all team members and remind them of upcoming deadlines. • Be the main point of contact for the clients Peter Cole and Mike Groeneweg. • Keep a list of potential problems (risk register) and update it. • Spot issues early before they become big problems. • Escalate issues to lecturer and in charge immediately • Make sure the team meets all project requirements. • Check that deliverables are submitted on time and in the correct format. • Help resolve conflicts between team members. • Make sure everyone has what they need to do their work. • Step in to help any role if someone is falling behind.
Business Analyst	Lim Yu Jun Felixia	• Understanding requirements and record down • Turn client needs into simple "user stories" • Write clear rules • Manage any changes or new requests from clients

		• Help translate between business needs and technical tasks • Work with the tester to make sure everything is tested • Write user manuals from a beginner's perspective
Technical Architect/ Developer	Roy Yoo Hojin, Gu QiYu	• Build the admin dashboard to see all QR codes (Show statistics: number of scans, last scan time, locations on a map) • Build the redirect system: direct user to make sure they scan the qr code using the mobile application • Store all scan data into database with consolidated/comprehensive report
Documentation	Alvin Rui Long Yeow,	• Develop QR code generator system • Document minutes for meeting • Collaborate with tester on her key responsibilities
Tester	Vanessa Tan	• Develop QR code generator system including specified features (expiry, redirect, geoloc etc) • Testing to ensure that generated QR codes and mobile application is functional. • Verifying the QR code authentication, redirection and expiry timeout. • Identifying bugs and features and recommending security measures and improvements.

		• Help to resolve issues and ensure system reliability, security and performance.

Our group's name: VAFPQR

Our goals

What is our group trying to accomplish?

1. Build a working system by the end of the teaching period
2. Make sure QR code and management platform is accessible and working
3. Able to work with Android

Our ground rules

- When will we attempt to meet (what time, how often, how long)?

  Meet twice every week

- Where will our meetings take place?

  - Thursday 7-8pm / 8-9pm
  - Friday 7-8pm / 8-9pm

- How do we inform each other when we cannot be there or are running late? How else can we communicate and keep others up to date?

  1. Notes and minutes taken every meeting
  2. Inform at least 3 hours earlier if unable to attend meeting with reasons.
  3. Take initiative to follow up with groupmates if you miss out any information.
  4. If any conflict votes for an agreement.

- How do we deal with lateness to meetings? What does "on time" mean?

  On time meeting time as discussed. 10min delay accepted unless informed in advance otherwise considered late.

- How do we deal with members who don't participate enough, participate too much, or distract the group from its task?

2 warnings if still no change inform lecturer.

- How are we going to make decisions?

  1. Discuss with team make sure everyone in on par and agrees to it
  2. Any **conflicts**, vote and follow majority
  3. Try to settle as soon as possible.

- What will we do if a group member's work doesn't meet our standards?

  1. Give feedback and suggestion on how we can improve together.
  2. Give support and help.
  3. Open to communication.

13.4 Appendix D – Glossary of terms

Term	**Definition**
QR Code	Quick Response Code
JWT	JSON Web Token
API	Application Programming Interface
UAT	User Acceptance Testing
GPS	Global Positioning System

13.5 Appendix E – Agenda and minutes of all client meetings

Chairperson: Patricia

Secretary: Felixia

<u>Agenda</u>

Item No.	**Description of Item**	**Introduced by**	**Approx. time for item**

1.	User validation and authentication requirements	Team	10 minutes
2.	Security and encryption necessities	Team	5 minutes
3.	User and system process workflow	Team	10 minutes
4.	QR code scanning and validation	Team	15 minutes
5.	Future steps and confirmation of project requirements	Team	5 minutes
Total time		45 minutes	

Meeting minutes

Meeting date: 26/5/26

Meeting commenced at: 1.05 pm

Meeting concluded at: 1.55 pm

Members present:

1. Vanessa Tan 35235647
2. Patricia Orenza 35075567
3. Lim Yu Jun Felixia 35544256
4. Roy Hojin Yoo 34845454

1. Authentication and validation

Authentication based on User ID was said to be suitable for the current phase of the project.

Action to be taken	Person/s responsible	Action to be completed by
Explore more on the user ID authentication approach.	Roy Yoo Hojin and Gu Qiyu	3/6/26

2. Authentication validation

Client stated that the solution does not necessarily have to be a dedicated QR scanning application, browser, OS-based and other existing scanners will be further explored.

Action to be taken	Person/s responsible	Action to be completed by
QR validation methods investigation	Roy Yoo Hojin and Gu Qiyu	3/6/26

3. Encryption and security

Encryption of data relating to QR codes and validation processes were addressed but the definitive implementation strategy is yet to be decided.

Action to be taken	Person/s responsible	Action to be completed by
Research on different types of encryptions methods for QR data and determine what is most ideal.	Vanessa Tan, Alvin Rui Long Yeow	3/6/26

4. User workflow and process workflow

Our proposed workflow was reviewed and accepted as a baseline structure.

Action to be taken	Person/s responsible	Action to be completed by
Enhance and document final system process	Lim Yu Jun Felixia	20/5/26

5. Documentation

The outcomes of meeting and project decisions will be recorded and stored for tracking of project.

Action to be taken	Person/s responsible	Action to be completed by
Keep track and document meeting minutes	Alvin Rui Long Yeow	20/5/26

6. Testing preparation

Requirements regarding testing were discussed briefly to make sure that the system's functionality aligns with the expectations before committing.

Action to be taken	Person/s responsible	Action to be completed by
Understand and prepare potential test cases and review the system's functionality.	Vanessa Tan	20/5/26

13.6 Appendix F – Agenda and minutes of all supervisor meetings

Chairperson: Patricia

Secretary: Felixia

<u>Week 2</u>

<u>Agenda</u>

Item No.	Description of Item	Introduced by	Approx. time for item
1.	Get to know more about team members	Team	10 minutes
2.	Understand the roles available	Team	10 minutes
Total time		20 minutes	

<u>Meeting minutes</u>

Meeting date: 13/5/26

Meeting commenced at: 10.15 am

Meeting concluded at: 10.40 am

Members present:

1. Vanessa Tan 35235647
2. Alvin Yeow 35475756
3. Patricia Orenza 35075567
4. Lim Yu Jun Felixia 35544256
5. Gu Qiyu 35139306
6. Roy Hojin Yoo 34845454

1. Organisation of team and mapping skills

We were just getting to know more about each other, understanding each other's skills and majors.

Action to be taken	Person/s responsible	Action to be completed by
Examining individual strengths to ensure balanced work distribution (Writing, documentation, technical)	All members	13/5/26
Allocating team roles (Project manager, business analyst, programmer, tester)	All members	13/5/26

2. Operational criteria and communication

We discussed and planned on ways to communicate with each other and when does everyone's schedule align for further meetings.

Action to be taken	Person/s responsible	Action to be completed by
Establishing a collaborative digital workspace (eg. Teams, OneDrive, Google drive)	Patricia (Project Manager)	**13/5/26**

Discussion and planning of weekly meeting time	All members	**13/5/26**

Chairperson: Patricia

Secretary: Felixia

Week 3

Agenda

Item No.	Description of Item	Introduced by	Approx. time for item
1.	Discussion for preparation of meeting with client	Team	10 minutes
2.	More specific delegation of duties to members	Team	20 minutes
3.	Completion of Team Charter	Team	20 Minutes
Total Time		50 Minutes	

Meeting minutes

Meeting date: 20/5/26

Meeting commenced at: 10.25 am

Meeting concluded at: 10.50 am

Members present:

1. Vanessa Tan 35235647
2. Alvin Yeow 35475756
3. Patricia Orenza 35075567
4. Lim Yu Jun Felixia 35544256
5. Roy Hojin Yoo 34845454

1. Client Preparation and Internal task delegation

Discussed what each member should do to prepare for the meeting with client

Action to be taken	Person/s responsible	Action to be completed by
Team discussion to prepare for meeting with client	All members	20/5/26

More specific delegation of tasks to each member to ensure a smoother workflow

Action to be taken	Person/s responsible	Action to be completed by
More specific delegation of duties to members	All members	20/5/26

2. Team completion of Team Charter

Discussion among team members to complete Team Charter

Action to be taken	Person/s responsible	Action to be completed by
Completion of Team Charter	All members	24/5/26

<u>Week 4</u>

<u>Agenda</u>

Item No.	Description of Item	Introduced by	Approx. time for item
1.	Discuss and share what we found out after research on existing QR systems	Team	20 minutes
2.	Additional preparation for client meeting	Team	15 minutes
3.	Brainstorming session for non-functional components of the project	Team	10 minutes
Total time		45 minutes	

Meeting minutes

Meeting date: 26/5/26

Meeting commenced at: 10.30 am

Meeting concluded at: 11 am

Members present:

1. Vanessa Tan 35235647
2. Lim Yu Jun Felixia 35544256
3. Roy Hojin Yoo 34845454

1. Updates before client meeting

After further research and discussions, we came up with more resources and questions to affirm our curiosity about the client's expectations.

Action to be taken	Person/s responsible	Action to be completed by
Update questions to confirm with the client.	All members	26/5/26
Create a draft process flow chart to let the client know our outline of the project.	All members	26/5/26

Week 5

Agenda

Item No.	Description of Item	Introduced by	Approx. time for item
1.	Confirming what software and tools we will be using for the QR code scanning and validation.	Roy, Qiyu	15 minutes
2.	Figuring out what security measures can be implemented with the software used	Vanessa, Alvin	15 minutes

3.	Come up with a more detailed structure of process to maximise time	Felixia, Patricia	15 minutes
4.	Update each other on individual process	Team	10 minutes
5.	Developing testing approach for system verification	Vanessa, Alvin	10 minutes
Total time		65 minutes	

Meeting minutes

Meeting date: 3/6/26

Meeting commenced at: 10.45 am

Meeting concluded at: 11.25 am

Members present:

1. Vanessa Tan 35235647
2. Lim Yu Jun Felixia 35544256
3. Gu Qiyu 35139306
4. Roy Hojin Yoo 34845454

1. Tools and software for QR system

Confirmation of software and tools that will be used for this project is discussed and further explanation of why and how they work.

Action to be taken	Person/s responsible	Action to be completed by
Proceed with development utilizing the tools and software	Roy, Qiyu	Ongoing

2. Security measures

Security protocols such as encryption methods relating to QR were reviewed

Action to be taken	Person/s responsible	Action to be completed by
Implementation plan for security and encryption methods	Vanessa, Alvin	Ongoing

3. System and workflow process structure

A more detailed process structure was discussed to ensure that workflow is efficient and equal.

Action to be taken	Person/s responsible	Action to be completed by
Follow and implement the updated system workflow.	Felixia, Patricia	Ongoing

4. Progress updates

Each team members to update their personal progress accordingly to roles assigned.

Action to be taken	Person/s responsible	Action to be completed by
Continue in developing accordingly to the roles assigned	All members	Ongoing